

# All Parametric Survival Models

METIMMOX-1 Economic Evaluation

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## Table of contents

|          |                                                           |           |
|----------|-----------------------------------------------------------|-----------|
| <b>1</b> | <b>Overview</b>                                           | <b>2</b>  |
| 1.1      | Model Specification . . . . .                             | 2         |
| 1.2      | Distributions Tested . . . . .                            | 2         |
| <b>2</b> | <b>Model Selection Summary</b>                            | <b>3</b>  |
| 2.1      | Overall Survival (OS) Models . . . . .                    | 3         |
| 2.2      | Progression-Free Survival (PFS) Models . . . . .          | 3         |
| <b>3</b> | <b>Overall Survival (OS) Model Coefficients</b>           | <b>4</b>  |
| 3.1      | Gamma (Best Fit) . . . . .                                | 4         |
| 3.2      | Exponential . . . . .                                     | 4         |
| 3.3      | Generalized F . . . . .                                   | 5         |
| 3.4      | Generalized Gamma . . . . .                               | 5         |
| 3.5      | Gompertz . . . . .                                        | 5         |
| 3.6      | Log-logistic . . . . .                                    | 6         |
| 3.7      | Log-normal . . . . .                                      | 6         |
| 3.8      | Weibull (AFT) . . . . .                                   | 6         |
| 3.9      | Weibull (PH) . . . . .                                    | 7         |
| <b>4</b> | <b>Progression-Free Survival (PFS) Model Coefficients</b> | <b>8</b>  |
| 4.1      | Gamma (Best Fit) . . . . .                                | 8         |
| 4.2      | Exponential . . . . .                                     | 8         |
| 4.3      | Generalized F . . . . .                                   | 8         |
| 4.4      | Generalized Gamma . . . . .                               | 9         |
| 4.5      | Gompertz . . . . .                                        | 9         |
| 4.6      | Log-logistic . . . . .                                    | 9         |
| 4.7      | Log-normal . . . . .                                      | 10        |
| 4.8      | Weibull (AFT) . . . . .                                   | 10        |
| 4.9      | Weibull (PH) . . . . .                                    | 10        |
| <b>5</b> | <b>Interpretation Guide</b>                               | <b>11</b> |
| 5.1      | Distribution-Specific Parameters . . . . .                | 11        |
| 5.2      | Coefficient Interpretation . . . . .                      | 11        |
| 5.3      | Biomarker Interaction Terms . . . . .                     | 11        |
| <b>6</b> | <b>Summary</b>                                            | <b>11</b> |

# 1 Overview

This technical document presents the regression coefficients for **all parametric survival models** fitted in the METIMMOX-1 economic evaluation. While the primary analysis uses the best-fitting model (selected by AIC), this document provides full transparency by showing coefficients for all distributions tested.

## 1.1 Model Specification

All models use the following covariate structure (full model with age and sex adjustment):

**Overall Survival (OS):**

$\text{Surv(OSwk, Death)} \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp}*\text{Rx} + \text{tlr}*\text{Rx} + \text{tmb\_braf}*\text{Rx}$

**Progression-Free Survival (PFS):**

$\text{Surv(PFSwk, Progression)} \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp}*\text{Rx} + \text{tlr}*\text{Rx} + \text{tmb\_braf}*\text{Rx}$

## 1.2 Distributions Tested

Nine parametric distributions were fitted to each outcome:

1. **Exponential** - Constant hazard rate
2. **Weibull (AFT)** - Accelerated failure time parameterization
3. **Weibull (PH)** - Proportional hazards parameterization
4. **Log-logistic** - Allows for non-monotonic hazard
5. **Log-normal** - Log-transformed normal distribution for survival times
6. **Gamma** - Two-parameter distribution, flexible shape
7. **Gompertz** - Hazard increases exponentially with time
8. **Generalized Gamma** - Three-parameter family including Weibull, gamma, and log-normal as special cases
9. **Generalized F** - Four-parameter family, most flexible

## 2 Model Selection Summary

### 2.1 Overall Survival (OS) Models

Table 1: Overall Survival model fit comparison (sorted by AIC)

| Distribution      | AIC    | BIC    | Delta AIC | Rank | Selected |
|-------------------|--------|--------|-----------|------|----------|
| Gamma             | 622.56 | 646.47 | 0.00      | 1.0  | ***      |
| Weibull (AFT)     | 623.44 | 647.36 | 0.88      | 2.5  |          |
| Weibull (PH)      | 623.44 | 647.36 | 0.88      | 2.5  |          |
| Log-logistic      | 623.68 | 647.60 | 1.12      | 4.0  |          |
| Log-normal        | 624.09 | 648.01 | 1.53      | 5.0  |          |
| Generalized Gamma | 624.55 | 650.65 | 1.99      | 6.0  |          |
| Generalized F     | 626.56 | 654.82 | 4.00      | 7.0  |          |
| Gompertz          | 633.09 | 657.01 | 10.53     | 8.0  |          |
| Exponential       | 654.14 | 675.88 | 31.58     | 9.0  |          |

### 2.2 Progression-Free Survival (PFS) Models

Table 2: Progression-Free Survival model fit comparison (sorted by AIC)

| Distribution      | AIC    | BIC    | Delta AIC | Rank | Selected |
|-------------------|--------|--------|-----------|------|----------|
| Gamma             | 453.43 | 477.35 | 0.00      | 1.0  | ***      |
| Log-normal        | 453.46 | 477.38 | 0.03      | 2.0  |          |
| Weibull (AFT)     | 454.19 | 478.11 | 0.76      | 3.5  |          |
| Weibull (PH)      | 454.19 | 478.11 | 0.76      | 3.5  |          |
| Log-logistic      | 455.16 | 479.08 | 1.73      | 5.0  |          |
| Generalized Gamma | 455.28 | 481.37 | 1.85      | 6.0  |          |
| Generalized F     | 457.28 | 485.55 | 3.85      | 7.0  |          |
| Gompertz          | 467.57 | 491.49 | 14.14     | 8.0  |          |
| Exponential       | 483.52 | 505.26 | 30.09     | 9.0  |          |

### 3 Overall Survival (OS) Model Coefficients

**Best-fitting distribution:** gamma (AIC: 622.56 )

The following tables present the regression coefficients for each fitted OS model. Parameters include:

- **Distribution-specific parameters** (e.g., shape, scale, rate)
- **Age** - Effect per year of age
- **Sex** - Effect of male vs female
- **Rx** - Main treatment effect (experimental vs control)
- **Biomarker:Rx interactions** - Treatment effect modification by biomarker status

#### 3.1 Gamma (Best Fit)

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| shape                        | 1.2264   | 0.1821 | 3.4090   |
| rate                         | -2.8997  | 0.6161 | 0.0550   |
| Age (per year)               | 0.0015   | 0.0082 | 1.0015   |
| Sex (male vs female)         | 0.1442   | 0.1482 | 1.1551   |
| RxExperimental arm           | -0.2348  | 0.2852 | 0.7907   |
| crp1                         | -0.3621  | 0.3169 | 0.6962   |
| tlr1                         | -0.8044  | 0.2509 | 0.4473   |
| tmb_braf1                    | -0.0890  | 0.2210 | 0.9148   |
| RxExperimental arm:crp1      | 0.0207   | 0.3913 | 1.0209   |
| RxExperimental arm:tlr1      | 0.3953   | 0.3295 | 1.4849   |
| RxExperimental arm:tmb_braf1 | -0.0946  | 0.3018 | 0.9098   |

#### 3.2 Exponential

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| rate                         | -4.4248  | 1.1022 | 0.0120   |
| Age (per year)               | 0.0063   | 0.0156 | 1.0064   |
| Sex (male vs female)         | 0.1835   | 0.2830 | 1.2014   |
| RxExperimental arm           | -0.1916  | 0.5253 | 0.8256   |
| crp1                         | -0.5534  | 0.6427 | 0.5750   |
| tlr1                         | -0.9110  | 0.4699 | 0.4021   |
| tmb_braf1                    | -0.1924  | 0.4242 | 0.8250   |
| RxExperimental arm:crp1      | 0.1371   | 0.7793 | 1.1470   |
| RxExperimental arm:tlr1      | 0.3861   | 0.6193 | 1.4712   |
| RxExperimental arm:tmb_braf1 | -0.1050  | 0.5789 | 0.9003   |

### 3.3 Generalized F

| Parameter                    | Estimate | SE      | exp(Est) |
|------------------------------|----------|---------|----------|
| mu                           | 4.1157   | 0.6323  | 61.2949  |
| sigma                        | -0.6103  | 0.1367  | 0.5432   |
| Q                            | 0.5270   | 0.4282  | 1.6938   |
| P                            | -7.1738  | 65.3299 | 0.0008   |
| Age (per year)               | -0.0014  | 0.0082  | 0.9986   |
| Sex (male vs female)         | -0.1447  | 0.1492  | 0.8653   |
| RxExperimental arm           | 0.2322   | 0.2875  | 1.2613   |
| crp1                         | 0.3559   | 0.3180  | 1.4275   |
| tlr1                         | 0.8046   | 0.2519  | 2.2358   |
| tmb_braf1                    | 0.0880   | 0.2243  | 1.0920   |
| RxExperimental arm:crp1      | -0.0153  | 0.3961  | 0.9848   |
| RxExperimental arm:tlr1      | -0.3924  | 0.3319  | 0.6754   |
| RxExperimental arm:tmb_braf1 | 0.0996   | 0.3025  | 1.1047   |

### 3.4 Generalized Gamma

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| mu                           | 4.1285   | 0.6310 | 62.0816  |
| sigma                        | -0.6097  | 0.1365 | 0.5435   |
| Q                            | 0.5269   | 0.4277 | 1.6936   |
| Age (per year)               | -0.0016  | 0.0082 | 0.9984   |
| Sex (male vs female)         | -0.1461  | 0.1492 | 0.8641   |
| RxExperimental arm           | 0.2282   | 0.2875 | 1.2563   |
| crp1                         | 0.3510   | 0.3177 | 1.4205   |
| tlr1                         | 0.8048   | 0.2520 | 2.2363   |
| tmb_braf1                    | 0.0845   | 0.2243 | 1.0882   |
| RxExperimental arm:crp1      | -0.0088  | 0.3959 | 0.9912   |
| RxExperimental arm:tlr1      | -0.3942  | 0.3320 | 0.6742   |
| RxExperimental arm:tmb_braf1 | 0.1041   | 0.3025 | 1.1097   |

### 3.5 Gompertz

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| shape                        | 0.0137   | 0.0026 | 1.0138   |
| rate                         | -5.5282  | 1.1991 | 0.0040   |
| Age (per year)               | 0.0126   | 0.0170 | 1.0127   |
| Sex (male vs female)         | 0.3678   | 0.2983 | 1.4446   |
| RxExperimental arm           | -0.3428  | 0.5268 | 0.7098   |
| crp1                         | -1.1291  | 0.6666 | 0.3233   |
| tlr1                         | -1.4740  | 0.4923 | 0.2290   |
| tmb_braf1                    | -0.1956  | 0.4296 | 0.8224   |
| RxExperimental arm:crp1      | 0.4749   | 0.8020 | 1.6079   |
| RxExperimental arm:tlr1      | 0.7020   | 0.6346 | 2.0178   |
| RxExperimental arm:tmb_braf1 | -0.0821  | 0.5975 | 0.9212   |

### 3.6 Log-logistic

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| shape                        | 1.0880   | 0.1125 | 2.9683   |
| scale                        | 3.6647   | 0.6313 | 39.0442  |
| Age (per year)               | 0.0028   | 0.0084 | 1.0028   |
| Sex (male vs female)         | -0.1517  | 0.1501 | 0.8592   |
| RxExperimental arm           | 0.2843   | 0.3371 | 1.3288   |
| crp1                         | 0.2045   | 0.3493 | 1.2269   |
| tlr1                         | 0.8812   | 0.2840 | 2.4137   |
| tmb_braf1                    | 0.0838   | 0.2482 | 1.0874   |
| RxExperimental arm:crp1      | 0.1383   | 0.4134 | 1.1484   |
| RxExperimental arm:tlr1      | -0.4512  | 0.3539 | 0.6368   |
| RxExperimental arm:tmb_braf1 | 0.1421   | 0.3165 | 1.1527   |

### 3.7 Log-normal

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| meanlog                      | 3.9003   | 0.6356 | 49.4154  |
| sdlog                        | -0.5154  | 0.0969 | 0.5973   |
| Age (per year)               | -0.0013  | 0.0083 | 0.9987   |
| Sex (male vs female)         | -0.1283  | 0.1548 | 0.8796   |
| RxExperimental arm           | 0.2676   | 0.3244 | 1.3068   |
| crp1                         | 0.3716   | 0.3345 | 1.4501   |
| tlr1                         | 0.7722   | 0.2727 | 2.1645   |
| tmb_braf1                    | 0.1527   | 0.2373 | 1.1650   |
| RxExperimental arm:crp1      | 0.0144   | 0.4121 | 1.0145   |
| RxExperimental arm:tlr1      | -0.3129  | 0.3608 | 0.7313   |
| RxExperimental arm:tmb_braf1 | 0.0806   | 0.3190 | 1.0839   |

### 3.8 Weibull (AFT)

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| shape                        | 0.7281   | 0.1090 | 2.0711   |
| scale                        | 4.5338   | 0.5431 | 93.1093  |
| Age (per year)               | -0.0051  | 0.0081 | 0.9949   |
| Sex (male vs female)         | -0.1654  | 0.1429 | 0.8475   |
| RxExperimental arm           | 0.2082   | 0.2536 | 1.2315   |
| crp1                         | 0.4164   | 0.3091 | 1.5164   |
| tlr1                         | 0.7730   | 0.2295 | 2.1663   |
| tmb_braf1                    | 0.0673   | 0.2081 | 1.0696   |
| RxExperimental arm:crp1      | -0.0990  | 0.3815 | 0.9058   |
| RxExperimental arm:tlr1      | -0.3977  | 0.3021 | 0.6719   |
| RxExperimental arm:tmb_braf1 | 0.0578   | 0.2888 | 1.0595   |

### 3.9 Weibull (PH)

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| shape                        | 0.7281   | 0.1090 | 2.0711   |
| scale                        | -9.3901  | 1.5795 | 0.0001   |
| Age (per year)               | 0.0105   | 0.0168 | 1.0106   |
| Sex (male vs female)         | 0.3426   | 0.2988 | 1.4086   |
| RxExperimental arm           | -0.4313  | 0.5275 | 0.6497   |
| crp1                         | -0.8624  | 0.6443 | 0.4222   |
| tlr1                         | -1.6010  | 0.4921 | 0.2017   |
| tmb_braf1                    | -0.1394  | 0.4303 | 0.8699   |
| RxExperimental arm:crp1      | 0.2049   | 0.7904 | 1.2275   |
| RxExperimental arm:tlr1      | 0.8237   | 0.6317 | 2.2789   |
| RxExperimental arm:tmb_braf1 | -0.1197  | 0.5978 | 0.8872   |

## 4 Progression-Free Survival (PFS) Model Coefficients

**Best-fitting distribution:** gamma (AIC: 453.43 )

The following tables present the regression coefficients for each fitted PFS model.

### 4.1 Gamma (Best Fit)

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| shape                        | 1.2423   | 0.1901 | 3.4637   |
| rate                         | -1.8724  | 0.6325 | 0.1538   |
| Age (per year)               | -0.0008  | 0.0084 | 0.9992   |
| Sex (male vs female)         | 0.1330   | 0.1653 | 1.1423   |
| RxExperimental arm           | 0.4228   | 0.3072 | 1.5262   |
| crp1                         | 0.0663   | 0.4254 | 1.0685   |
| tlr1                         | -0.9773  | 0.2717 | 0.3763   |
| tmb_braf1                    | -0.1039  | 0.2536 | 0.9013   |
| RxExperimental arm:crp1      | -0.6418  | 0.4911 | 0.5263   |
| RxExperimental arm:tlr1      | -0.2142  | 0.3486 | 0.8072   |
| RxExperimental arm:tmb_braf1 | -0.3863  | 0.3409 | 0.6796   |

### 4.2 Exponential

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| rate                         | -3.0223  | 1.1468 | 0.0487   |
| Age (per year)               | -0.0026  | 0.0161 | 0.9975   |
| Sex (male vs female)         | 0.1749   | 0.3161 | 1.1911   |
| RxExperimental arm           | 0.3956   | 0.5700 | 1.4853   |
| crp1                         | -0.0452  | 0.7953 | 0.9558   |
| tlr1                         | -1.0805  | 0.5136 | 0.3394   |
| tmb_braf1                    | -0.2308  | 0.4812 | 0.7939   |
| RxExperimental arm:crp1      | -0.5242  | 0.9260 | 0.5920   |
| RxExperimental arm:tlr1      | -0.1622  | 0.6622 | 0.8503   |
| RxExperimental arm:tmb_braf1 | -0.4092  | 0.6555 | 0.6642   |

### 4.3 Generalized F

| Parameter                    | Estimate | SE      | exp(Est) |
|------------------------------|----------|---------|----------|
| mu                           | 3.0808   | 0.6539  | 21.7760  |
| sigma                        | -0.5735  | 0.1450  | 0.5635   |
| Q                            | 0.2873   | 0.6748  | 1.3328   |
| P                            | -8.5840  | 66.2147 | 0.0002   |
| Age (per year)               | 0.0001   | 0.0088  | 1.0001   |
| Sex (male vs female)         | -0.1380  | 0.1651  | 0.8711   |
| RxExperimental arm           | -0.3981  | 0.3283  | 0.6716   |
| crp1                         | 0.0162   | 0.5002  | 1.0163   |
| tlr1                         | 0.9721   | 0.2842  | 2.6435   |
| tmb_braf1                    | 0.0878   | 0.2655  | 1.0917   |
| RxExperimental arm:crp1      | 0.5755   | 0.5443  | 1.7780   |
| RxExperimental arm:tlr1      | 0.2522   | 0.3831  | 1.2869   |
| RxExperimental arm:tmb_braf1 | 0.3657   | 0.3436  | 1.4415   |

#### 4.4 Generalized Gamma

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| mu                           | 3.0812   | 0.6541 | 21.7847  |
| sigma                        | -0.5734  | 0.1448 | 0.5636   |
| Q                            | 0.2852   | 0.6756 | 1.3301   |
| Age (per year)               | 0.0000   | 0.0089 | 1.0000   |
| Sex (male vs female)         | -0.1379  | 0.1651 | 0.8712   |
| RxExperimental arm           | -0.3974  | 0.3284 | 0.6721   |
| crp1                         | 0.0169   | 0.5005 | 1.0170   |
| tlr1                         | 0.9720   | 0.2843 | 2.6432   |
| tmb_braf1                    | 0.0874   | 0.2655 | 1.0914   |
| RxExperimental arm:crp1      | 0.5741   | 0.5447 | 1.7756   |
| RxExperimental arm:tlr1      | 0.2529   | 0.3833 | 1.2878   |
| RxExperimental arm:tmb_braf1 | 0.3655   | 0.3436 | 1.4412   |

#### 4.5 Gompertz

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| shape                        | 0.0223   | 0.0052 | 1.0225   |
| rate                         | -3.4252  | 1.1475 | 0.0325   |
| Age (per year)               | -0.0023  | 0.0166 | 0.9977   |
| Sex (male vs female)         | 0.2329   | 0.3290 | 1.2623   |
| RxExperimental arm           | 0.6939   | 0.5688 | 2.0016   |
| crp1                         | 0.3533   | 0.8147 | 1.4238   |
| tlr1                         | -1.6909  | 0.5483 | 0.1844   |
| tmb_braf1                    | -0.2949  | 0.4879 | 0.7446   |
| RxExperimental arm:crp1      | -1.3147  | 1.0200 | 0.2685   |
| RxExperimental arm:tlr1      | -0.2892  | 0.6760 | 0.7489   |
| RxExperimental arm:tmb_braf1 | -1.0307  | 0.7592 | 0.3568   |

#### 4.6 Log-logistic

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| shape                        | 1.0833   | 0.1187 | 2.9545   |
| scale                        | 3.2498   | 0.7145 | 25.7854  |
| Age (per year)               | -0.0034  | 0.0090 | 0.9966   |
| Sex (male vs female)         | -0.1807  | 0.1679 | 0.8347   |
| RxExperimental arm           | -0.4030  | 0.3671 | 0.6683   |
| crp1                         | 0.0133   | 0.4021 | 1.0134   |
| tlr1                         | 1.0309   | 0.3232 | 2.8036   |
| tmb_braf1                    | 0.0161   | 0.2996 | 1.0163   |
| RxExperimental arm:crp1      | 0.5770   | 0.4699 | 1.7807   |
| RxExperimental arm:tlr1      | 0.2448   | 0.3996 | 1.2774   |
| RxExperimental arm:tmb_braf1 | 0.4047   | 0.3661 | 1.4988   |

## 4.7 Log-normal

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| meanlog                      | 3.0492   | 0.6934 | 21.0982  |
| sdlog                        | -0.5406  | 0.1029 | 0.5824   |
| Age (per year)               | -0.0008  | 0.0089 | 0.9992   |
| Sex (male vs female)         | -0.1413  | 0.1636 | 0.8683   |
| RxExperimental arm           | -0.3740  | 0.3369 | 0.6880   |
| crp1                         | 0.1227   | 0.4559 | 1.1305   |
| tlr1                         | 0.9463   | 0.2895 | 2.5761   |
| tmb_braf1                    | 0.0689   | 0.2661 | 1.0713   |
| RxExperimental arm:crp1      | 0.4790   | 0.5195 | 1.6145   |
| RxExperimental arm:tlr1      | 0.3088   | 0.3796 | 1.3618   |
| RxExperimental arm:tmb_braf1 | 0.3505   | 0.3412 | 1.4198   |

## 4.8 Weibull (AFT)

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| shape                        | 0.7437   | 0.1139 | 2.1038   |
| scale                        | 3.2164   | 0.5353 | 24.9377  |
| Age (per year)               | 0.0015   | 0.0080 | 1.0015   |
| Sex (male vs female)         | -0.1076  | 0.1647 | 0.8980   |
| RxExperimental arm           | -0.4631  | 0.2751 | 0.6293   |
| crp1                         | -0.1966  | 0.3853 | 0.8215   |
| tlr1                         | 0.9559   | 0.2475 | 2.6011   |
| tmb_braf1                    | 0.1323   | 0.2349 | 1.1414   |
| RxExperimental arm:crp1      | 0.7180   | 0.4642 | 2.0504   |
| RxExperimental arm:tlr1      | 0.1692   | 0.3137 | 1.1844   |
| RxExperimental arm:tmb_braf1 | 0.4547   | 0.3454 | 1.5757   |

## 4.9 Weibull (PH)

| Parameter                    | Estimate | SE     | exp(Est) |
|------------------------------|----------|--------|----------|
| shape                        | 0.7437   | 0.1139 | 2.1038   |
| scale                        | -6.7666  | 1.3964 | 0.0012   |
| Age (per year)               | -0.0032  | 0.0169 | 0.9968   |
| Sex (male vs female)         | 0.2264   | 0.3463 | 1.2541   |
| RxExperimental arm           | 0.9743   | 0.5936 | 2.6492   |
| crp1                         | 0.4137   | 0.8152 | 1.5123   |
| tlr1                         | -2.0111  | 0.5566 | 0.1338   |
| tmb_braf1                    | -0.2783  | 0.4940 | 0.7571   |
| RxExperimental arm:crp1      | -1.5106  | 0.9982 | 0.2208   |
| RxExperimental arm:tlr1      | -0.3560  | 0.6607 | 0.7005   |
| RxExperimental arm:tmb_braf1 | -0.9566  | 0.7384 | 0.3842   |

## 5 Interpretation Guide

### 5.1 Distribution-Specific Parameters

Different parametric distributions have different ancillary parameters:

| Distribution      | Ancillary Parameters |
|-------------------|----------------------|
| Exponential       | rate                 |
| Weibull (AFT)     | shape, scale         |
| Weibull (PH)      | shape, scale         |
| Log-logistic      | shape, scale         |
| Log-normal        | meanlog, sdlog       |
| Gamma             | shape, rate          |
| Gompertz          | shape, rate          |
| Generalized Gamma | mu, sigma, Q         |
| Generalized F     | mu, sigma, Q, P      |

### 5.2 Coefficient Interpretation

- **Estimate**: The coefficient on the log scale (for AFT models) or log-hazard scale (for PH models)
- **SE**: Standard error of the coefficient estimate
- **exp(Est)**: Exponentiated coefficient
  - For AFT models: time ratio (values >1 indicate longer survival)
  - For PH models: hazard ratio (values <1 indicate lower hazard/better survival)

### 5.3 Biomarker Interaction Terms

The biomarker:Rx interaction terms indicate treatment effect modification:

- **crp1:RxExperimental** - Effect of experimental treatment in CRP-positive patients (relative to CRP-negative)
- **tlr1:RxExperimental** - Effect of experimental treatment in TLR-positive patients
- **tmb\_braf1:RxExperimental** - Effect of experimental treatment in TMB\_BRAF-positive patients

A significant interaction suggests the biomarker predicts differential treatment benefit.

## 6 Summary

This document presents the full set of parametric survival model coefficients for the METIMMOX-1 economic evaluation. The best-fitting models (selected by AIC) are:

| Outcome                         | Best Distribution | AIC    |
|---------------------------------|-------------------|--------|
| Overall Survival (OS)           | Gamma             | 622.56 |
| Progression-Free Survival (PFS) | Gamma             | 453.43 |

All model coefficients are provided for transparency and to support sensitivity analyses exploring alternative distributional assumptions.

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